MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 08** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in **Test Paper:** Periodic Test 08

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. The mean of the squares of first five natural numbers is: [1]

(A) 11 (B) 10 (C) 12 (D) 15

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 11

Justification: The squares of first five natural numbers are $1^2, 2^2, 3^2, 4^2, 5^2 = 1, 4, 9, 16, 25$.

$$\text{Mean} = \frac{1+4+9+16+25}{5} = \frac{55}{5} = \mathbf{11}.$$

$$\boxed{\text{Mean} = \mathbf{11}}$$

Q2. In the formula Median $= l + \left(\frac{N/2 - cf}{f}\right) \times h$, cf is the cumulative frequency of the: [1]

(A) median class (B) class preceding the median class
(B) class succeeding the median class (D) modal class

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) class preceding the median class

Justification: In the grouped median formula, cf specifically denotes the cumulative frequency of the class interval immediately preceding the median class.

Q3. If the ratio of mean and median of a certain data is 2 : 3, then the ratio of its mode and mean is: [1]

(A) 3 : 2 (B) 5 : 2 (C) 2 : 5 (D) 7 : 2

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 5 : 2

Justification: Let Mean $= 2k$ and Median $= 3k$.
Using the empirical relationship:

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean}) = 3(3k) - 2(2k) = 9k - 4k = 5k.$$

$$\frac{\text{Mode}}{\text{Mean}} = \frac{5k}{2k} = \frac{5}{2} = \mathbf{5 : 2}.$$

Ratio of Mode to Mean = **5 : 2**

Q4. For the following grouped data, the modal class is: [1]

Class	10 – 15	15 – 20	20 – 25	25 – 30	30 – 35
Frequency	30	35	20	40	15

(A) 15 – 20 (B) 20 – 25 (C) 25 – 30 (D) 10 – 15

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (C) 25 – 30

Justification: The highest frequency in the distribution is 40, which belongs to the class interval **25 – 30**.

Q5. If the mean of observations $x, x+3, x+5, x+7, x+10$ is 9, then the mean of last three observations is: [1]

(A) 10.33 (B) 11.33 (C) 12 (D) 10

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 11.33

Justification:

$$\text{Mean} = \frac{x + (x+3) + (x+5) + (x+7) + (x+10)}{5} = \frac{5x + 25}{5} = x + 5$$

Given Mean $= 9 \implies x + 5 = 9 \implies x = 4$.
The last three observations are: $x + 5 = 9$, $x + 7 = 11$, and $x + 10 = 14$.

$$\text{Mean of last 3} = \frac{9 + 11 + 14}{3} = \frac{34}{3} = \mathbf{11.33} \quad \left(\text{or } 11\frac{1}{3}\right).$$

Mean of last three observations = **11.33**

Q6. **Assertion (A):** In finding the mean of grouped data by step-deviation method, the class size h must be uniform across all classes.
Reason (R): The step-deviation variable $u_i = \dfrac{x_i - a}{h}$ requires a single common division factor h. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: The standard step-deviation formula $\bar{x} = a + \left(\dfrac{\sum f_i u_i}{N}\right) \times h$ is derived on the assumption that a single common width h divides all class deviations $x_i - a$. If class widths vary, a single factor h cannot be factored out. Thus, both statements are true and Reason (R) correctly explains Assertion (A).

SECTION B: Very Short Answer Type — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. If Mode − Median = 12, find the value of Median − Mean. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formula Relation [1 Mark]
From empirical relationship:

$$\text{Mode} = 3\text{Median} - 2\text{Mean} \implies \text{Mode} - \text{Median} = 2(\text{Median} - \text{Mean})$$

Step 2: Substitution & Value Finding [1 Mark]
Given Mode − Median = 12:

$$12 = 2(\text{Median} - \text{Mean}) \implies \text{Median} - \text{Mean} = \frac{12}{2} = \mathbf{6}.$$

$$\boxed{\text{Median} - \text{Mean} = \mathbf{6}}$$

Q8. If the mean of 10 observations is 25 and that of another 15 observations is 35, find the combined mean of all 25 observations. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Total Sum of Observations [1 Mark]
Sum of first 10 observations: $\sum x_1 = 10 \times 25 = 250$.
Sum of next 15 observations: $\sum x_2 = 15 \times 35 = 525$.
Total Sum $= 250 + 525 = 775$.

Step 2: Combined Mean Calculation [1 Mark]

$$\bar{x}_{\text{combined}} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{775}{10 + 15} = \frac{775}{25} = \mathbf{31}.$$

$$\boxed{\text{Combined Mean} = \mathbf{31}}$$

SECTION C: Short Answer Type — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. Find the mean number of defective components per batch from the given records: **[3]**

Defects	$0-4$	$4-8$	$8-12$	$12-16$	$16-20$
Batches	3	7	12	10	8

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Direct Method Calculation Table [1.5 Marks]

Defects	Batches (f_i)	Class Mark (x_i)	f_ix_i
$0-4$	3	2	6
$4-8$	7	6	42
$8-12$	12	10	120
$12-16$	10	14	140
$16-20$	8	18	144
Total	$\sum f_i = 40$		$\sum f_ix_i = 452$

Step 2: Formula Declaration & Substitution [1 Mark]

$$\bar{x} = \frac{\sum f_ix_i}{\sum f_i} = \frac{452}{40} = \frac{113}{10} = 11.3$$

Step 3: Final Boxed Answer [0.5 Mark]

$$\boxed{\text{Mean Number of Defective Components} = \mathbf{11.3}}$$

Q10. Calculate the mode of the following distribution of scores: **[3]**

Class	$20-30$	$30-40$	$40-50$	$50-60$	$60-70$
Frequency	25	40	42	33	10

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Modal Class & Parameter Identification [1 Mark]

The maximum frequency is 42, which lies in the class interval $\mathbf{40-50}$.
$\therefore$ Modal Class $= 40-50$.
Lower limit $(l) = 40$, class width $(h) = 10$.
$f_1 = 42$ (modal frequency), $f_0 = 40$ (preceding frequency), $f_2 = 33$ (succeeding frequency).

Step 2: Mode Formula & Substitution [1 Mark]

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{Mode} = 40 + \left(\frac{42-40}{2(42)-40-33}\right) \times 10 = 40 + \left(\frac{2}{84-73}\right) \times 10 = 40 + \frac{20}{11}$$

Step 3: Calculation & Final Boxed Answer [1 Mark]

$$\text{Mode} = 40 + 1.818... \approx \mathbf{41.82} \quad \left(\text{or } 41\frac{9}{11}\right)$$

$$\boxed{\text{Mode of Scores} = \mathbf{41.82} \quad \left(\text{or } \mathbf{41\frac{9}{11}}\right)}$$

SECTION D: Long Answer Type

MATHS MANTHAN | **(2 × 5 = 10 Marks)**

Q11. The median of the following frequency distribution is 28.5. Find the values of x and y, if the total frequency is 60: **[5]**

Class Interval	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$	$50-60$
Frequency	5	x	20	15	y	5

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Cumulative Frequency Table & Equation (i) [1.5 Marks]

Class Interval	Frequency (f_i)	Cumulative Frequency (cf)
$0-10$	5	5
$10-20$	x	$5+x$
$20-30$	20	$25+x$
$30-40$	15	$40+x$
$40-50$	y	$40+x+y$
$50-60$	5	$45+x+y$
Total	$N=60$	

Given total frequency $N=60$:

$$45+x+y=60 \implies x+y=60-45 \implies x+y=15 \quad \text{— (i)}$$

Step 2: Median Class Identification & Parameters [1 Mark]

Given Median $= 28.5$, which lies in the class interval **$20-30$**.

$\therefore$ Median Class $= 20-30$.

Lower limit $(l)=20$, $f=20$, $h=10$, $\dfrac{N}{2}=30$, $cf=5+x$ (preceding cf).

Step 3: Solving for x via Median Formula [1.5 Marks]

$$\text{Median} = l+\left(\frac{\frac{N}{2}-cf}{f}\right)\times h$$

$$28.5 = 20+\left(\frac{30-(5+x)}{20}\right)\times 10$$

$$8.5=\frac{25-x}{2} \implies 17=25-x \implies x=25-17=\mathbf{8}.$$

Step 4: Finding y & Final Result [1 Mark]

Substituting $x=8$ into Equation (i):

$$8+y=15 \implies y=15-8=\mathbf{7}.$$

$$\boxed{x=\mathbf{8},\quad y=\mathbf{7}}$$

Q12. Compute the Mean, Median, and Mode for the following frequency distribution: **[5]**

Class Interval	$10-20$	$20-30$	$30-40$	$40-50$	$50-60$	$60-70$
Frequency	4	8	10	12	6	4

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Part 1: Mean Calculation [1.5 Marks]

Class	f_i	x_i	cf	f_ix_i
$10-20$	4	15	4	60
$20-30$	8	25	12	200
$30-40$	10	35	22	350
$40-50$	12	45	34	540
$50-60$	6	55	40	330
$60-70$	4	65	44	260
Total	$\sum f_i = 44$			$\sum f_ix_i = 1740$

$$\bar{x} = \frac{\sum f_ix_i}{\sum f_i} = \frac{1740}{44} = \frac{435}{11} \approx \mathbf{39.55}.$$

Part 2: Median Calculation [1.5 Marks]

$N = 44 \implies \dfrac{N}{2} = 22$. Median class is $\mathbf{30-40}$ ($l = 30, cf = 12, f = 10, h = 10$).

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h = 30 + \left(\frac{22-12}{10}\right) \times 10 = 30 + 10 = \mathbf{40}.$$

Part 3: Mode Calculation [1.5 Marks]

Maximum frequency $= 12 \implies$ Modal class is $\mathbf{40-50}$ ($l = 40, f_1 = 12, f_0 = 10, f_2 = 6, h = 10$).

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h = 40 + \left(\frac{12-10}{24-10-6}\right) \times 10 = 40 + 2.5 = \mathbf{42.5}.$$

$\boxed{\text{Mean} = \mathbf{39.55}, \quad \text{Median} = \mathbf{40}, \quad \text{Mode} = \mathbf{42.5}}$ [0.5 Mark]

SECTION E: Case Study Based Question

MATHS MANTHAN | $(1 \times 4 = 4$ Marks)

Q13. CASE STUDY: MUNICIPAL PUBLIC HEALTH BLOOD PRESSURE SCREENING INITIATIVE [4 Marks]

A metropolitan civic body conducted an adult cardiovascular health checkup camp, measuring the systolic blood pressure (mm Hg) of 80 visiting citizens.

Systolic BP (mm Hg)	$100-110$	$110-120$	$120-130$	$130-140$	$140-150$	$150-160$
Number of Citizens	8	14	25	18	10	5

(i) Find the median systolic blood pressure of the screened citizens. [2]

(ii) Identify the modal blood pressure class and state its lower boundary. [1]

(iii) How many citizens were identified with stage-1 or stage-2 hypertension (BP $\geq$ 140 mm Hg)? [1]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Subpart (i): Median Systolic Blood Pressure [2 Marks]

Total citizens $N = 80 \implies \dfrac{N}{2} = 40$.

Cumulative frequencies: $100-110 : 8$; $110-120 : 22$; $120-130 : 47$; $130-140 : 65$; $140-150 : 75$; $150-160 : 80$.

Cumulative frequency just greater than 40 is 47, corresponding to median class **120 – 130**.

$l = 120, cf = 22, f = 25, h = 10$.

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h \text{[1 Mark]}$$

$$\text{Median} = 120 + \left(\frac{40-22}{25}\right) \times 10 = 120 + \frac{180}{25} = 120 + 7.2 = \mathbf{127.2} \text{ mm Hg [1 Mark]}$$

$$\boxed{\text{Median Systolic BP} = \mathbf{127.2} \text{ mm Hg}}$$

Subpart (ii): Modal Class & Lower Boundary [1 Mark]

The maximum frequency is 25, which belongs to class **120 – 130** mm Hg.

Lower boundary $(l) = 120$ mm Hg.

$$\boxed{\text{Modal Class} = \mathbf{120-130} \text{ mm Hg}, \quad \text{Lower Boundary} = \mathbf{120} \text{ mm Hg}}$$

Subpart (iii): Citizens with BP ≥ 140 mm Hg [1 Mark]

Citizens in intervals $[140-150]$ and $[150-160]$:

$$\text{Total Citizens} = 10 + 5 = \mathbf{15}.$$

$$\boxed{\text{Citizens with BP } \geq 140 \text{ mm Hg} = \mathbf{15}}$$

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